



Weapon-LLM

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Background

- VirusTotal reports processing approximately 1.8million samples every day. [1]
- To remain effective in such environments, **malware detection systems must resilient against adversarial manipulation.**
- With assistance of **LLM**, we could generate **unseen variants of malware** to train malware detection to make it robust.

[1] VirusTotal. VirusTotal : Analyse suspicious files, domains, IP s and URLs to detect malware and other breaches, automatically share them with the security community., 2025.

Manipulation in code level

악성 코드 데이터로 LLM 을 **fine-tuning** 시켜 LLM 이 더 다양한 멀웨어를 생성하게 한다. 이때, 생성한 코드는 반드시 **evasive & executable** 해야 한다.

Challenges:

Fine-tuning 할 만큼의 데이터가 없음.

publicly available 한 악성 코드 데이터가 존재하지 않음. (.exe 형태로는 존재)

VX-Underground 에서 적은 양의 소스 코드 갯수가 접근 가능했는데, 빌드가 불가능.

Manipulation in PE level

코드 레벨이 아닌, **PE 파일 자체에 perturbation** 을 더해서 디텍션을 **evasion** 한다

PE 파일에 .reloc 섹션에 dummy data 를 추가한 후, probability score 가 감소하였음.

0.980822 → 0.020476. Benign

```
# benign-looking printable strings
benign_strings = [
    b"This application was developed by Microsoft.\x00",
    b"Compiled using Visual Studio.\x00",
    b"All rights reserved.\x00",
    b"Version 1.0.0.1\x00"
```

```
]
```

Related works - Adversarial Malware

Mab-Malware

- Mab-Malware uses the Reinforcement Learning and a set of actions to use. They define the Macro-action and Micro-actions, which has different range of the changes in the PE file.

Type	Abbr	Name	Description
Macro	OA	Overlay Append	Appends benign contents at the end of a binary
	SP	Section Append	Appends random bytes to the unused space at the end of a section.
	SA	Section Add	Adds a new section with benign contents.
	SR	Section Rename	Change the section name to a name in benign binaries.
	RC	Remove Certificate	Zero out the signed certificate of a binary.
	RD	Remove Debug	Zero out the debug information in a binary.
	BC	Break Checksum	Zero out the checksum value in the optional header.
	CR	Code Randomization	Replace instruction sequence with semantically equivalent one
Micro	OA1	Overlay Append 1 Byte	Appends 1 byte at the end of a binary
	SP1	Section Append 1 Byte	Appends 1 byte to the unused space at the end of a section.
	SA1	Section Add 1 Byte	Adds a new section with 1 byte content.
	SR1	Section Rename 1 Byte	Change 1 byte of a section name.
	CP1	Code Section Append 1 Byte	Appends 1 byte to the unused space at the end of the code section.

[2] Wei Song et al. MAB- Malware : A reinforcement learning framework for black-box generation of adversarial malware. In Proceedings of the ACM on Asia Conference on Computer and Communications Security (Asia CCS), 2022.

Attack Strategy for PE mutation

어택 타겟인 EMBER 는 트리 기반의 모델,
모델의 내부에 접근 가능해도 어떤 조건이 **adversarial** 한 지 알 수 없음.

RL 프레임워크 또한 **explicit evasive behaviour** 를 보장하지 않음.

-> 어떤 **Feature** 조건이 왜 **evasion** 가능한 지 설명할 수 있는 존



THE FUTURE
WE CREATE

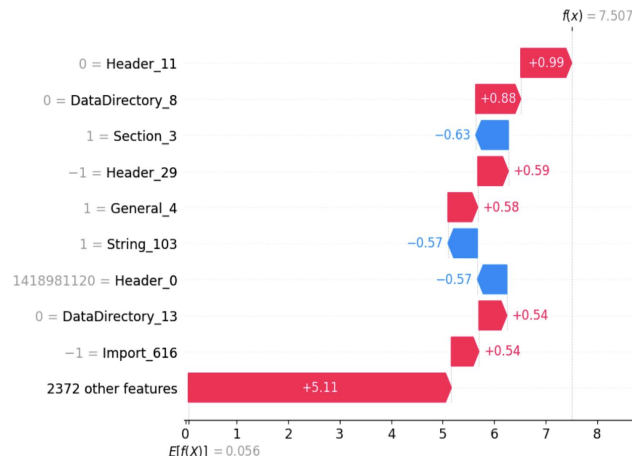
Preliminary - Shapley Additive Explanations (SHAP)

- **SHAP**

SHAP (SHapley Additive exPlanations) is a game theoretic approach to explain the output of any machine learning model

With the help of SAHP module, it is possible to select which features and values to modify, along with the explanation of their contribution

[3] Scott M . Lundberg et al. From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, 2020



LLM Guided Feature-to-Executable

- Generating adversarial features with Shapley Additive Explanations (SHAP)
- Feature-to-Executable Generation with Prompt

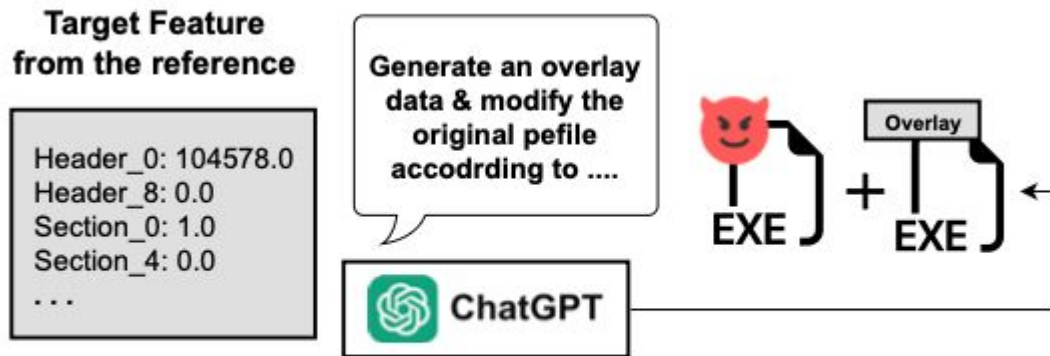


Fig. 1. Overview of Feature-to-Executable

Generating adversarial features with Shapley Additive Explanations (SHAP)

To flip a prediction, we adopt a greedy perturbation strategy at each step, **the feature with the largest positive contribution is replaced** with a benign candidate value from a reference set that minimizes its SHAP score.

Algorithm 1: SHAP-Guided Greedy Perturbation

Input: Sample x , SHAP values $\{\phi_i(x)\}$, reference set R

Output: Perturbed sample x'

$x' \leftarrow x$;

while $\sum_i \phi_i(x') \geq 0$ **do**

$j \leftarrow \arg \max_i \phi_i(x')$;

 // largest malicious contribution

$v^* \leftarrow \arg \min_{v \in R_j} \phi_j(x'_{[j] \leftarrow v})$;

$x'_{[j]} \leftarrow v^*$;

 recompute $\{\phi_i(x')\}$;

return x'

EWHA,
THE FUTURE
WE CREATE

Feature-to-Executable Generation with Prompt

We leverage LLMs to translate perturbed feature vectors into valid PE binaries or byte-histogram features, the LLM is prompted to generate overlay code that adjusts byte counts without corrupting the file structure.

***Prompt.** Please generate code that constructs the overlay Obyte so that the executable meets the following condition: feature j : target value v .*

Target Feature
from the reference

```
Header_0: 104578.0  
Header_8: 0.0  
Section_0: 1.0  
Section_4: 0.0  
...
```

Generate an overlay
data & modify the
original pefile
accodrding to

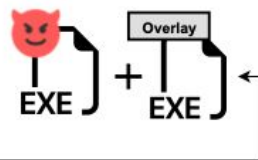


Fig. 1. Overview of Feature-to-Executable

Evaluation

- **Dataset** - 10,000 randomly selected malware from VirusShare
- **Model** - EMBER
- **Attack Success Rate**

TABLE I
ATTACK SUCCESS RATE OF FEATURE AND GENERATED EXECUTABLE

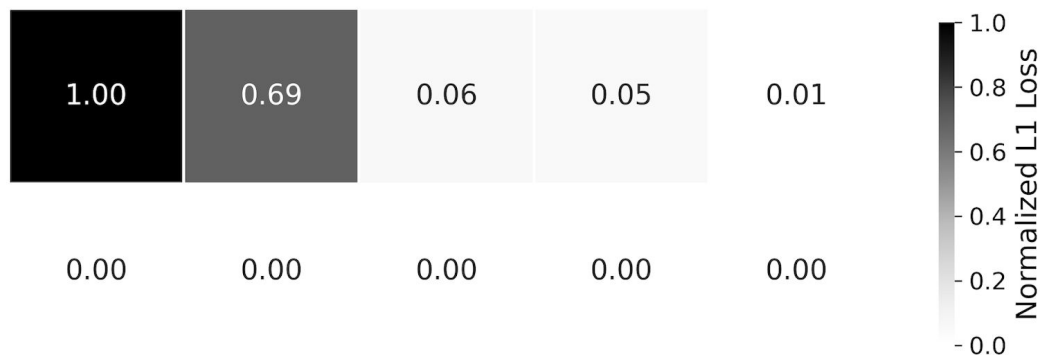
Adversarial Attack	Attack Success Rate (%)
MAB-malware	71.39
Our Method (feature)	87.20
Our Method (executable)	55.14

Evaluation

Robustness of feature-to-executable translation:

As shown in the figure, most bytes are reconstructed with minimal error, but some target values deviate significantly when the required perturbation is too large to realize without excessive overhead. These reconstruction gaps directly reduce ASR when perturbations are translated into executable form.

Top 10 Loss of Byte Histogram Features Reconstruction



Subsequent Experiment

exe 로 변환하는 게 불가능한 featureHash 들

예를 들어, Header_characteristics_0 는 전체 헤더의 캐릭터리스틱을 리스트로 만든 것을, feature Hash 에 통과시켜 10 차원의 hash 로 만든 후, 이때 0번째 인덱스의 값

```
"DataDirectory_CERTIFICATE_size_0": "7000.0",  
"Section_size_hash_32": "-512.0",  
"Section_RX_0": "1.0",  
"Header_characteristics_0": "-1.0",  
"Header_minor_op_system_0": "0.0",  
"Section_entropy_hash_32": "-0.04021612",  
"Header_subsystem_8": "0.0",
```

Subsequent Experiment

LLM 의 비중을 더 키우기 위해 : **LLM As a decision Maker**

LLM 에게 여러가지 액션을 제공하고, 그 액션이 **evasion** 에 미친 영향을 평가하고, 로그를 LLM 에게 제공하여 LLM이 액션을 선택함.

Decision Maker for Action	Content Pool	ASR - EMBER
LLM	Benign (Assembly Dataset)	15%
LLM	Misclassified sample	17%

Subsequent Experiment

LLM 의 출력과 pe에 반영될 때, 또 디텍션이 보는 feature 들 사이의 gap

LLM 이 출력할 수 있는 것	출력이 실제 PE 파일에 반영될 때	디텍션이 PE 에서 추출하는 feature
문자열	코드 → 컴파일을 거쳐 .obj 파일이 된 후 pe 파일에 반영.	EMBER feature pe 파일의 feature 들 + Hash

Conclusion

LLM 이 갖고 있는 능력과,

Malware Variants Generation, Mutation, Evasion 에 필요한 능력은 서로 다름.